

ARYA COLLEGE OF ENGINEERING

I MIDTERM EXAMINATION 2025-26(V Sem.)

5CS4-03|5AID4-03|5CCS4-03:OPERATING SYSTEM

(BRANCH: CSE|AI & DS(Sec-C,D,E,F,G,H))

Time: - 2 Hr.

Max. Marks: 60

SECTION-I (Attempt All)

	CO	BTL	MARKS
(a) Define Operating System	CO-1	L-1	(3)
(b) Define mutual exclusion.	CO-1	L-2	(3)
(c) Explain fragmentation.	CO-2	L-2	(3)
(d) What do you mean by Memory Management?	CO-2	L-1	(3)
(e) Differentiate between process and threads	CO-1	L-1	(3)

SECTION-II (Attempt any four)

(a) Define the Semaphore with an example.	CO-1	L-1	(6)
A System has RAM = 32 MB, process size = 8 MB. Each process has CPU burst = 12ms and I/O burst = 18ms. Find: (i) Number of processes in memory (ii) CPU utilization (iii) I/O utilization (iv) Effect of multiprogramming on CPU utilization	CO-2	L-1	(6)
(c) Differentiate Single & Multiple Threading	CO-1	L-6	(6)
(d) Define different process states in detail with a neat and clean diagram.	CO-1	L-1	(6)
(e) Describe Peterson's Solution for the critical section problem..	CO-1	L-1	(6)
(f) Describe contiguous memory allocation & its type.	CO-1	L-1	(6)

SECTION-III (Attempt any Two)

SECTION II (Part B) (10.5)

	(a) Define interprocess communication. Its methods with advantages and disadvantages. For the following set of process, find the average waiting time and turnaround time using. Gantt chart for (i) SJF (ii) Priority scheduling process <table border="1" data-bbox="250 1348 1008 1684"> <thead> <tr> <th>Process</th><th>AT</th><th>BT</th><th>Priority (1=Highest)</th></tr> </thead> <tbody> <tr> <td>P1</td><td>0</td><td>10</td><td>3</td></tr> <tr> <td>P2</td><td>1</td><td>1</td><td>1</td></tr> <tr> <td>P3</td><td>2</td><td>2</td><td>4</td></tr> <tr> <td>P4</td><td>3</td><td>1</td><td>5</td></tr> <tr> <td>P5</td><td>4</td><td>5</td><td>2</td></tr> </tbody> </table>	Process	AT	BT	Priority (1=Highest)	P1	0	10	3	P2	1	1	1	P3	2	2	4	P4	3	1	5	P5	4	5	2	CO-1	L-1	(10.5)
Process	AT	BT	Priority (1=Highest)																									
P1	0	10	3																									
P2	1	1	1																									
P3	2	2	4																									
P4	3	1	5																									
P5	4	5	2																									
Q3	(b)	CO-1	L-3	(10.5)																								
	(i) Define first fit, next fit, best fit, and worst fit with a suitable example. (ii) Requests from processes are 300K, 25K, 125K, and 50K, respectively. The above requests could be satisfied with proper solution diagram. 1. Best Fit but not First Fit 2. First Fit but not Best Fit 3. Both 4. None	CO-2	L-3	(10.5)																								

ARYA COLLEGE OF ENGINEERING

I MIDTERM EXAMINATION 2025-26(V Sem.)

5CS4-05|5CCS4-05|5CAI4-055IT4-05|5AID4-05: ANALYSIS OF ALGORITHM

(BRANCH: CSE|CYBS|CS(AI)|IT|AI & DS)

Max. Marks: 60

Time: - 2 Hr.

SECTION-I (Attempt All)

	CO	BTL	MARKS
(a) Difference between dynamic programming and the Greedy method.	CO-2	L-3	(3)
(b) Sort the given elements using quick sort: 13, 28, 35, 65, 45, 90, 25.	CO-1	L-2	(3)
Q1 (c) What is a Minimum Spanning Tree? Name any two algorithms used to find the Minimum Spanning Tree.	CO-2	L-2	(3)
(d) Find the longest common subsequence from ABCDBCDCCDDACD and BCDCDAD.	CO-2	L-3	(3)
(e) Find out the complexity of the given program. A() {i=1, S=1; while(S<=n) {i++; S=S+1; print("Ravi");}}	CO-1	L-5	(3)

SECTION-II (Attempt any four)

(a) Contrast between 3 Asymptotic notations and define the use of the Objective of Asymptotic Notations.	CO-1	L-2	(6)
(b) Find the optimal merge pattern for instance 14, 9, 12, 2, 5, 1, 5, 7, 3, 4, 6.	CO-2	L-3	(6)
(c) Using Prim's and Kruskal's Algorithm. Find out the minimum cost for a given graph.	CO-2	L-2	(6)
Q2 (d) Find out the solution generated by job sequencing. When n=7 with the following profit and deadline. Profit(p1, p2, ..., p7) = {3, 5, 20, 18, 1, 6, 30}, Deadline(d1, d2, ..., d7) = {1, 3, 4, 3, 2, 1, 2}	CO-2	L-5	(6)
(e) Solve the recurrence using the master theorem. T(n) = 8T(n/2) + n ² (ii) T(n) = T(n ^{1/2}) + C	CO-1	L-5	(6)
(f) Find a solution by a greedy approach of the following instance of the knapsack problem: - fractional w = {3, 10, 5, 6, 8, 4} and v = {15, 15, 30, 18, 16, 22} where capacity of knapsack is 20. Calculate the total profit.	CO-2	L-3	(6)

SECTION-III (Attempt any Two)

(a) Show all the steps of Strassen's matrix multiplication algorithm to multiply the following matrices. X = $\begin{bmatrix} 1 & 2 \\ 6 & 3 \end{bmatrix}$	CO-1	L-5	(10.5)
Y = $\begin{bmatrix} 5 & 7 \\ 3 & 1 \end{bmatrix}$			
Q3 (b) Find the optimal parenthesization of the matrix chain product whose sequence of dimensions is (6, 12, 6, 42, 7).	CO-2	L-2	(10.5)
(c) Differentiate b/w merge sort and quick sort with a suitable example.	CO-1	L-4	(10.5)

ARYA COLLEGE OF ENGINEERING

I MIDTERM EXAMINATION 2025-26(V Sem.)
SCS5-11|SIT5-11: WIRELESS COMMUNICATION
(BRANCH: CSE|IT)

Time: - 2 Hr.

Max marks: 60

SECTION-I (Attempt All)

	CO	BTL	MARKS
Q1 (a) Define the Cellular concept?	CO-2	L-3	(3)
(b) What is Doppler shift?	CO-1	L-2	(3)
(c) What do you mean by: (i) Coherence Bandwidth (ii) Doppler spread & coherence Time	CO-1	L-2	(3)
(d) What is Fading? Explain large-scale Fading.	CO-1	L-2	(3)
(e) What is the Scope and objective of wireless communication?	CO-1	L-2	(3)

SECTION-II (Attempt any four)

Q2 (a) What do you mean Two-Ray path model with a diagram?	CO-1	L-2	(6)
(b) Explain the different types of interference in the cellular concept.	CO-2	L-2	(6)
(c) Define free space path loss and link budget.	CO-1	L-3	(6)
(d) Explain the Microcell Zone concept with an example.	CO-2	L-2	(6)
(e) Discuss a) Cell splitting technique (b) Cell Sectoring technique	CO-2	L-2	(6)
(f) Explain the concept of the Handoff.	CO-2	L-2	(6)

SECTION-III (Attempt any Two)

SECTION-III (Attempt any two)

The power delay profile of the given channel is

Delays (micro Sec).	0	1	2	3	4	5
Fading in dB	-30	-20	-10	-10	-10	0

(a)

CO-1

L-3

(10.5)

Calculate the mean excess delay and RMS delay spread for the channel, and also calculate the 90% and 50% coherence bandwidth of the channel.

(b)

What is Fading? Explain Small and large-scale Fading.

CO-1

L-2

(10.5)

(c)

Differentiate between FDMA, TDMA, and CDMA. Also discuss their system capacity and the advantages and disadvantages of each technique.

CO-2

L-1

(10.5)

ARYA COLLEGE OF ENGINEERING
I MIDTERM EXAMINATION 2025-26(V Sem.)
5CS3-01|5CCS3-01: INFORMATION THEORY & CODING
(BRANCH: CSE|CYBS)

Max. Marks: 60

Time: - 2 Hr.

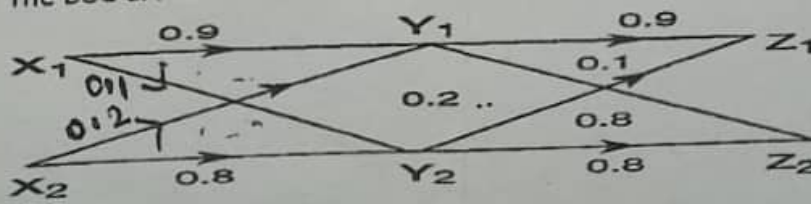
SECTION-I (Attempt All)

	CO	BTL	MARKS
Q1 (a) Draw a noiseless channel matrix with a channel diagram.	CO-1	L-3	(3)
(b) Define Source Coding along with its expression?	CO-1	L-2	(3)
(c) What do you mean by entropy?	CO-1	L-1	(3)
(d) Define Information. Prove that the following statement, 'The Sun rises from the east,' contains zero amount of information.	CO-1	L-1	(3)
(e) Explain the Kraft inequality.	CO-2	L-2	(3)

SECTION-II (Attempt any four)

Q2 (a) Construct a Shannon-Fano code for the following set of messages. Also find the efficiency of the code with $P(x_1)=0.3$, $P(x_2)=0.29$, $P(x_3)=0.16$, $P(x_4)=0.15$, $P(x_5)=0.1$	CO-2	L-3	(6)
(b) The probabilities of five possible outcomes of an experiment is given as $P(X_1)=1/2$, $P(X_2)=1/4$, $P(X_3)=1/8$, and $P(X_4)=1/16$. Determine the entropy and information rate if there are 16 outcomes per second.	CO-1	L-3	(6)
(c) A voice-grade channel of a telephone network has a bandwidth of 3.4 kHz. Calculate the information capacity of the telephone channel for a signal-to-noise ratio of 30dB	CO-1	L-3	(6)
(d) Explain any two properties of Mutual Information.	CO-1	L-2	(6)
(e) Define Lemple Zip coding. Generate Lemple Zip code for the given sequence. AABABBBABAABABBBBABBB	CO-2	L-1	(6)
(f) Show that the discrete binding channel $H(X,Y) = H(X)+H(Y)$	CO-1	L-3	(6)

SECTION-III (Attempt any Two)

Q3 (a) A DMS has seven symbols $x_1, x_2, x_3, x_4, x_5, x_6, x_7$ with $P(x_1)=0.3$, $P(x_2)=0.3$, $P(x_3)=0.1$, $P(x_4)=0.1$, $P(x_5)=0.08$, $P(x_6)=0.08$, $P(x_7)=0.04$. Construct the Huffman code and find its efficiency.	CO-2	L-3	(10.5)
(b) Channel capacity is given by $C = B \log_2 \left(1 + \frac{S}{N}\right)$ bits/sec. Prove the upper limit is given as $C_\infty = \lim_{S \rightarrow \infty} C = 1.44 \frac{S}{N}$	CO-2	L-2	(10.5)
(c) The BSC are connected in cascade as shown in the figure.  Determine the transition matrix for a discrete memoryless channel. Determine $P(z_1)$ and $P(z_2)$ if $p(x_1) = 0.6$ and $p(x_2) = 0.4$	CO-2	L-3	(10.5)

ARYA COLLEGE OF ENGINEERING

I MIDTERM EXAMINATION 2025-26(V Sem.)

5CS4-02|SCCS4-02|5CAI4-02|5IT4-02|5AID4-02: COMPILER DESIGN

(BRANCH: CSE|CYBS|CS(AI)|IT|AI & DS)

Time: - 2 Hr.

Max. Marks: 60

SECTION-I (Attempt All)

	CO	BLT	MARKS
Q1 (a) Define Left factoring with an example?	CO-2	L-2	(3)
(b) What is an Error Handler?	CO-1	L-2	(3)
(c) Differentiate between Bottom-up and Top-down parsing.	CO-2	L-3	(3)
(d) Differentiate between a compiler and an interpreter.	CO-1	L-2	(3)
(e) How would you explain the T-diagram of bootstrapping?	CO-1	L-4	(3)

SECTION-II (Attempt any four)

Find first and follow in a given grammar? Where #-Epsilon			
1.) $E \rightarrow TR$ 2.) $S \rightarrow AaAb/BbBa$			
(a) $R \rightarrow +TR \mid \#$ $T \rightarrow FY$ $Y \rightarrow *FY \mid \#$ $F \rightarrow (E) \mid i$	CO-2	L-2	(6)
(b) Remove Left Recursion from the given grammar if present: $E \rightarrow E+T \mid T$ $T \rightarrow T*F \mid F$ $F \rightarrow (E) \mid id$	CO-2	L-5	(6)
(c) Define Input Buffering in detail?	CO-1	L-2	(6)
(d) What is an LALR(1) grammar? Construct the LALR parsing table in the following grammar $S \rightarrow cC, C \rightarrow cC, C \rightarrow c/d$	CO-2	L-4	(6)
(e) How can you describe the following terms - i) EOF, begin pointer, and forward pointer buffer pair ii) Token, pattern, lexeme	CO-1	L-4	(6)
(f) Write a short note on : a) Lex compiler b) YACC compiler	CO-1	L-6	(6)

SECTION-III (Attempt any Two)

Q3 (a) Explain in detail about Recursive-Descent Parsing.	CO-2	L-2	(10.5)
(b) Explain the Phases of Compiler with an example in detail?	CO-1	L-2	(10.5)
(c) How would you explain different types of errors in compiler design?	CO-1	L-4	(10.5)

ARYA COLLEGE OF ENGINEERING

I MIDTERM EXAMINATION 2025-26(V Sem.)
5CS4-04|5CCS4-04|5CAI4-04|5IT4-04|5AID4-04:

COMPUTER GRAPHICS AND MULTIMEDIA

(BRANCH: CSE|CYBS|CS(AI)|IT|AI & DS)

Max. Marks: 60

Time: - 2 Hr.

SECTION-I (Attempt All)

	CO	BTL	MARKS
(a) Write two names of standard graphics software?	CO-1	L-3	(3)
(b) What is meant by resolution and Aspect ratio?	CO-1	L-3	(3)
(c) Define Graphics primitives with an example	CO-2	L-3	(3)
(d) What is the refresh rate in a 640*712 raster if pixels are accessed at the rate of 100 nanoseconds	CO-1	L-2	(3)
(e) Write four applications of computer graphics	CO-1	L-3	(3)

SECTION-II (Attempt any four)

(a) What steps are required to plot a line using DDA algorithm? Give an example	CO-2	L-3	(6)
(b) Explain and compare Raster Scan and Random Scan systems with neat diagrams.	CO-1	L-4	(6)
(c) Draw a circle having a radius = 10, using the mid-point circle drawing algorithm.	CO-2	L-4	(6)
(d) In a raster scan system with a resolution of 2560 * 2048. How many pixels could be accessed per second by a display controller that refreshes the screen at a rate of 60 frames per second? Calculate access time per pixel in the system.	CO-1	L-1	(6)
(e) Compare and construct the CRT and DVST with the help of a diagram.	CO-1	L-4	(6)
(f) Perform a 45 degree rotation of triangle a(0 0) b(1 1) c(5 2) about an arbitrary point(-1, -1)	CO-2	L-2	(6)

SECTION-III (Attempt any Two)

(a) Explain the following terms in the context of display devices 3. Flickering And Interlacing 4. Boundary Fill And Flood Fill Algorithm	CO-1,2	L-1	(10.5)
(b) Draw the line using Bresenham's line algorithm with end points (20,10) and (30,18) & what steps are required to plot a line by using Bresenham's Algorithm.	CO-2	L-4	(10.5)
(c) What is Transformation? Explain all the basic transformation techniques.	CO-3	L-4	(10.5)